

SpotSteer

DOCUMENTATION

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SpotSteer traffic light calculator

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Introduction

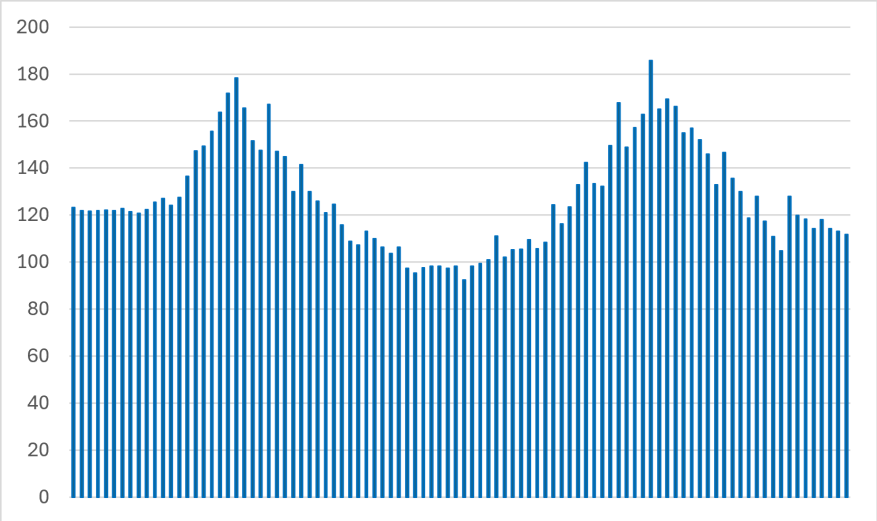
In June 2019, the EU Parliament passed Directive (EU) 2019/944, which granted citizens of the European Union the right to dynamic energy price contracts. This opens up a lot of new possibilities for the european consumer. Instead of relying on fixed energy contracts, they can take advantage of significant intraday price swings by changing their usage habits. In today's world of generally high energy prices, it is convenient to have the possibility of reducing the household's electricity bill. Therefore, we at SpotSteer decided to exploit this area and help families across the EU reduce their energy spending by assisting them in deciding when the right time is to do their laundry, charge their EV, heat up the water in their electric boilers, or run the dishwasher. Unlike our competitors, we decided to adopt a hybrid approach to deciding when the price is low and when it is high. In this paper, we describe the workings of this approach and show how it compares to some other strategies.

1 Assigning colors

In deciding which approach to adopt, we took into consideration 3 different strategies. All three of these strategies did not rely on a fixed price, as is done by some of our competitors. From the very beginning, we understood that each day is unique and that we cannot set one fixed price boundary for all days. That could create situations where, in one or two weeks, all time intervals would be red, meaning our application would be of no help to consumers. Below, we describe the 3 strategies we considered adopting. By discussing both, we will make a strong case for the one chosen by SpotSteer.

1.1 Day only focus

Our main task at SpotSteer is to help the citizens of the EU who decide to abandon their fixed energy contracts and go on the spot to determine when it is the right time to use their energy. The European system with day-ahead prices gives people great help in planning energy demanding activities in advance. However, when people look at tomorrow's price graph, see picture 1.1, it is easy to notice the price spikes in the morning and in the afternoon, and lower prices around midday. It is, however, challenging to determine when the barrier between low and high prices begins. Therefore, the first approach we considered was to



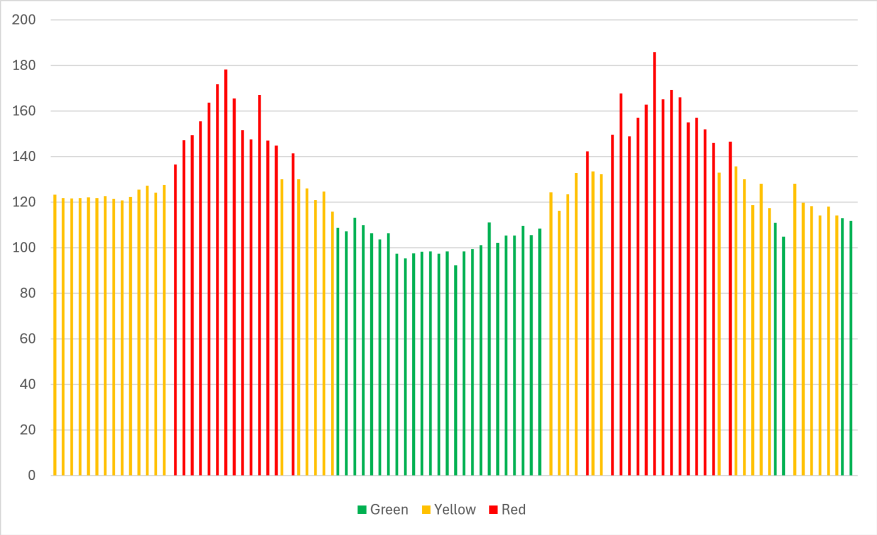
Pictures 1.1 Unmarked energy prices chart on the 5th of May 2026.

take the given day's prices and compute the 0.3 and 0.7 quantiles. So, we defined

$$\text{Color}_t = \begin{cases} \text{Green}, & P_t \leq q_{0.3}(p) \\ \text{Yellow}, & q_{0.3}(p) < P_t \leq q_{0.7}(p) \\ \text{Red}, & P_t > q_{0.7}(p) \end{cases},$$

where $q_{0.3}(p)$ and $q_{0.7}(p)$ represent the 0.3rd and 0.7th quantile of prices on a given day. using this algorithm, prices below the 0.3 quantile were considered cheap, and prices above the 0.7 quantile were considered expensive. Results of this

method can be seen in the picture 1.2. There, it can be clearly seen which prices are considered low, high, and which ones are neutral. More importantly, users can easily acknowledge when exactly the price breaks from being expensive to being cheap. Thanks to this color assignment, users could easily see when the right time



Pictures 1.2 One day based quantile graph for the 5th of May 2026.

is to use their electric appliances, plan their washing machine cycles, charge EVs or batteries, and more on a given day. The disadvantage of this approach was that it did not take into account potentially high price swings between given days. As can be seen in the picture 1.3, following the extremely negative prices in the first half of the week, green prices appear the next day even at 100 euros.



Pictures 1.3 One day based quantile graph for the week from the 30th of April to the 6th of May 2026.

Remark. In the picture 1.3 the energy price of -500 euros cannot go unnoticed. It is, however, not a data error. The time period on which we demonstrate our algorithm is the begging of the May 2026 in Germany. In this time period, a rather strange phenomenon of negative energy prices occured in not just Germany,

but accross the whole European Union. This sort of extreme situation is not common in the energy markets, however, at SpotSteer, we believe that algorithms should be made to withstand even this, as Nasim Taleb might put it, *extremistian* environment, and we we want to demonstrate how our startegy performs in those times.

We believe, however, that big price gaps between days should be reflected in the semaphore, and shortly after a period of very low energy prices, even negative ones, users should not be encouraged the next day to use the energy because 2 scenarios might happen. The first one is that after prolonged low prices on the spot market, it might happen that a phenomenon with a German nickname, *dunkelflaute*, hits the particular country for 1 day and spikes the energy prices. This spike might only be short term, and the question arises: should the users be encouraged to use the energy, or should they be more cautious and maybe wait for the prices to come down again. The second scenario is that a trend shift happens in the markets. It is not only a days spike that occurs. In both cases, we believe that users should be cautious and wait to see how it plays out and which of the scenarios happen. This line of thought brings us to the following section.

1.2 The history matters

The brief story at the end of the last section brings us to the question: how should we correctly navigate our customers during a day-to-day price shift in the energy markets. At SpotSteer, we decided not to advise users to use energy after this price increase, i.e., to create a method that would not encourage them to use electricity if a sudden and rapid shift in energy prices occurs. How can we help them determine whether it is only a one day spike or whether they are witnessing a bigger price shift in the markets. The first approach considered at SpotSteer was to simply extend the time frame. This would mean that instead of calculating price quantiles on a daily basis, we would take all prices from the last 5 or 7 days, calculate quantiles based on those values, and then assign colors. What this method would look like in practice can be seen in the picture 1.4. From this dataset, it can be clearly seen where the potential problem lies in using this method. In the first half of the week, when negative prices appeared, the semaphore behaved in accordance with expectations. However, as the week progressed and the prices started to behave as one might expect, the semaphore stayed red for the majority of the remaining week, marking only brief periods yellow and even fewer green. This kind of prolonged period of red/yellow time zones during a trend shift in the markets—is something we, as mentioned in the previous section, wanted to avoid. That is why we needed to find an approach that would, even in these uncommon times, react more appropriately to trend shifts.

1.3 Following the trend

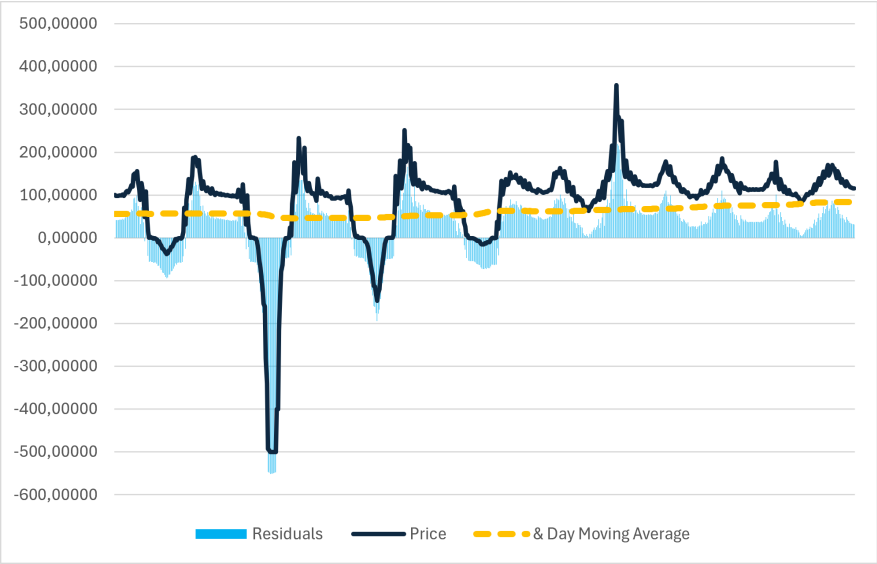
We would like to begin this section by briefly summing up the observations from the 2 previously mentioned methods. The key is that we want a method that takes into account trend shifts across days. Furthermore, we want it to react



Pictures 1.4 Seven day based quantile graph for the week from the 30th of April to the 6th of May 2026.

swiftly to these macro changes so that we do not witness multiple days of only *red* prices. For these demands, a simple 7 day moving average seemed like an excellent tool. Other time frames for the moving average were also considered, i.e., 3 day or 30 day moving average. However, we believe that 7 days represents the best tradeoff between the timeframe being too short or unnecessarily extensive.

In the next step, we calculated the residuals for prices from the 7 day moving average; see picture 1.5, which shows the residual values from the 30th of April to the 6th of May 2026. Formally, the residual at time t is defined as $r_t = P_t - MA_7(t)$, where P_t is the price at time t and $MA_7(t)$ represents the value of the 7 day moving average at time t. Then we calculated 0.3 and 0.7 quantiles for the residual



Pictures 1.5 Residuals chart for the week from the 30th of April to the 6th of May 2026.

data for the last week. Prices for which residuals were below the 0.3 quantile were assigned a green color, and prices with residuals above the 0.7 quantile were



Pictures 1.6 Seven day moving average based graph for the week from the 30th of April to the 6th of May 2026.

marked red; see picture 1.6. So,

$$\text{Color}_t = \begin{cases} \text{Green}, & r_t \leq q_{0.3}(r) \\ \text{Yellow}, & q_{0.3}(r) < r_t \leq q_{0.7}(r) \\ \text{Red}, & r_t > q_{0.7}(r) \end{cases},$$

where $q_{0.3}(r)$ and $q_{0.7}(r)$ represent the 0.3rd and 0.7th quantiles of residuals from the last 7 days. By using this approach, we are able to provide our customers with excellent navigation through energy prices, capturing both intra- and cross day shifts in today's volatile energy markets.

Conclusion

To briefly sum up this paper, the motivation of the team behind SpotSteer is to navigate European consumers in a highly volatile energy market, as so called spot contracts are becoming more and more popular. We acknowledge that the competition in this industry is very strong. However, we still tried to break into this market by using a different, more advanced approach to calculate the *good* and *bad* periods for energy consumption, in comparison to our competitors. Our journey, however, does not end here. We are still working on finding ways to improve our algorithm, so that we can bring the best results to our customers.